

CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Windows commands

C:\Users\Santhosh A>hostname

Santhosh A

C:\Users\Santhosh A>ipconfig /all

Windows IP Configuration

Host Name: Santhosh A

Primary Dns Suffix:

Node Type: Mixed

IP Routing Enabled.: No

WINS Proxy Enabled.: No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix .:

Description: VirtualBox Host-Only Ethernet Adapter

Physical Address.: 0A-00-27-00-00-0E

DHCP Enabled.: No

Autoconfiguration Enabled: Yes

Link-local IPv6 Address: fe80::aa4a:775:1de1:1757%14(Preferred)

IPv4 Address.: 192.168.56.1(Preferred)

Subnet Mask: 255.255.255.0

Default Gateway :

DHCPv6 IAID : 118095911

DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpip. : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. : F8-FE-5E-3A-70-56

DHCP Enabled. : Yes

Autoconfiguration Enabled . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected

Connection-specific DNS Suffix . :

Description : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. : FA-FE-5E-3A-70-55

DHCP Enabled. : No

Autoconfiguration Enabled . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description : Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. : F8-FE-5E-3A-70-55

DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::b810:29c9:55e9:3a65%17(Preferred)
IPv4 Address. : 10.83.96.83(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : 26 August 2026 14:38:14
Lease Expires : 26 August 2026 15:38:11
Default Gateway : 10.83.96.20
DHCP Server : 10.83.96.20
DHCPv6 IAID : 167312990
DHCPv6 Client DUID. : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F
DNS Servers : 10.83.96.20
NetBIOS over Tcpi. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : F8-FE-5E-3A-70-59
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek Gaming GbE Family Controller

Physical Address. : 2C-58-B9-DE-5F-5F

DHCP Enabled. : Yes

Autoconfiguration Enabled : Yes

C:\Users\Santhosh A>arp -a

Interface: 192.168.56.1 --- 0xe

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 10.83.96.83 --- 0x11

Internet Address	Physical Address	Type
10.83.96.20	4e-ec-85-ec-ff-15	dynamic
10.83.96.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

C:\Users\Santhosh A>netstat -r

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Interface List

14...0a 00 27 00 00 0eVirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56Microsoft Wi-Fi Direct Virtual Adapter
4...fa fe 5e 3a 70 55Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59Bluetooth Device (Personal Area Network)
2...2c 58 b9 de 5f 5fRealtek Gaming GbE Family Controller
1.....Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35
10.83.96.0	255.255.255.0		On-link	10.83.96.83	291
10.83.96.83	255.255.255.255		On-link	10.83.96.83	291
10.83.96.255	255.255.255.255		On-link	10.83.96.83	291
127.0.0.0	255.0.0.0		On-link	127.0.0.1	331
127.0.0.1	255.255.255.255		On-link	127.0.0.1	331
127.255.255.255	255.255.255.255		On-link	127.0.0.1	331
192.168.56.0	255.255.255.0		On-link	192.168.56.1	281
192.168.56.1	255.255.255.255		On-link	192.168.56.1	281
192.168.56.255	255.255.255.255		On-link	192.168.56.1	281
224.0.0.0	240.0.0.0		On-link	127.0.0.1	331
224.0.0.0	240.0.0.0		On-link	192.168.56.1	281
224.0.0.0	240.0.0.0		On-link	10.83.96.83	291
255.255.255.255	255.255.255.255		On-link	127.0.0.1	331
255.255.255.255	255.255.255.255		On-link	192.168.56.1	281

255.255.255.255 255.255.255.255 On-link 10.83.96.83 291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
14	281	fe80::/64		On-link
17	291	fe80::/64		On-link
14	281	fe80::aa4a:775:1de1:1757/128		
				On-link
17	291	fe80::b810:29c9:55e9:3a65/128		
				On-link
1	331	ff00::/8		On-link
14	281	ff00::/8		On-link
17	291	ff00::/8		On-link

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Persistent Routes:

None

C:\Users\Santhosh A>nslookup www.google.com

Server: UnKnown

Address: 10.83.96.20

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\Santhosh A>ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data: Reply

from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Santhosh A>route print

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Interface List

14...0a 00 27 00 00 0e .....VirtualBox Host-Only Ethernet Adapter
11...f8 fe 5e 3a 70 56 .....Microsoft Wi-Fi Direct Virtual Adapter
 4...fa fe 5e 3a 70 55 .....Microsoft Wi-Fi Direct Virtual Adapter #2
17...f8 fe 5e 3a 70 55 .....Intel(R) Wi-Fi 6E AX211 160MHz
19...f8 fe 5e 3a 70 59 .....Bluetooth Device (Personal Area Network)
 2...2c 58 b9 de 5f 5f .....Realtek Gaming GbE Family Controller
 1.....Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network Destination    Netmask          Gateway         Interface Metric
0.0.0.0                0.0.0.0    10.83.96.20    10.83.96.83  35
10.83.96.0            255.255.255.0      On-link      10.83.96.83  291
10.83.96.83          255.255.255.255      On-link      10.83.96.83  291
10.83.96.255        255.255.255.255      On-link      10.83.96.83  291
127.0.0.0            255.0.0.0      On-link      127.0.0.1  331
127.0.0.1          255.255.255.255      On-link      127.0.0.1  331
127.255.255.255    255.255.255.255      On-link      127.0.0.1  331
192.168.56.0        255.255.255.0      On-link     192.168.56.1  281
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192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	10.83.96.83	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281
255.255.255.255	255.255.255.255	On-link	10.83.96.83	291

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Persistent Routes:

None

IPv6 Route Table

=====

Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
14	281	fe80::/64		On-link
17	291	fe80::/64		On-link
14	281	fe80::aa4a:775:1de1:1757/128		
				On-link
17	291	fe80::b810:29c9:55e9:3a65/128		
				On-link
1	331	ff00::/8		On-link
14	281	ff00::/8		On-link
17	291	ff00::/8		On-link

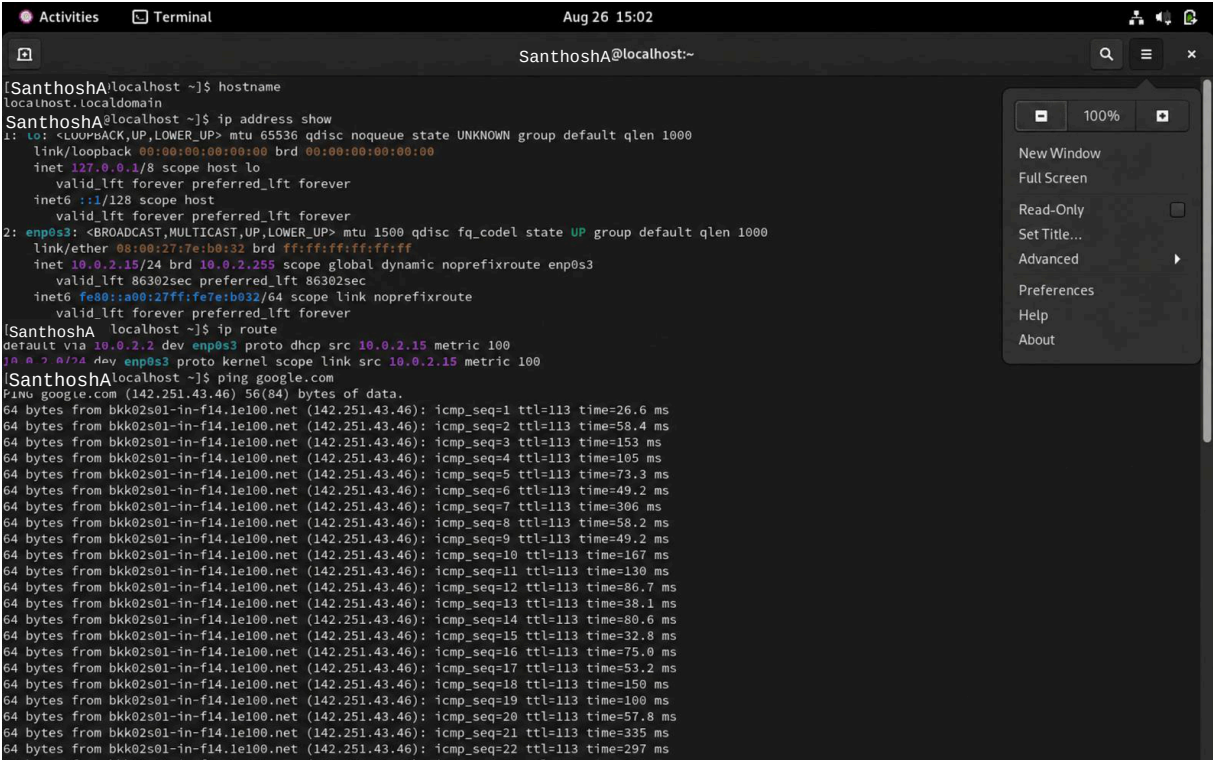
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Persistent Routes:

None

C:\Users\Santhosh A>tracert google.com

Linux commands



Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route